

Validation and Reliability Assessment of Autopsy Forensic Suite for Artifact Extraction

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Abstract — Forensic tools play a crucial role in digital investigations, but their reliability is rarely validated systematically. This paper aims to present a validation framework applied to an open-source toolkit named Autopsy forensic suite to measure its artifact extraction accuracy, metadata integrity, and repeatability. A controlled dataset was used to simulate realistic forensic evidence including deleted files, image metadata, and web artifacts. Results from Autopsy are compared against the manually verified data. Findings highlight areas where Autopsy performs reliably and where there are inconsistencies that appear. Additionally, the paper contributes to practical guidelines for small labs to implement lightweight validation procedures.

Keywords — Tool Validation, Digital Forensics, Artifact Extraction, Autopsy, The Sleuth Kit, NIST

I. INTRODUCTION

Digital forensic investigations depend on software tools to identify, extract, and interpret evidence from storage media such as hard drives. Courts and investigators rely on these results to make decisions. Hence, tool reliability and transparency are vital. However, open-source forensic tools often lack formal validation methods, leaving uncertainty regarding result accuracy and reproducibility.

This paper focuses on validating Autopsy (The Sleuth Kit) as a representative open-source forensic suite. The goal is to evaluate its performance in recovering deleted and existing files, maintaining metadata integrity, and providing consistent outputs across repeated runs. By developing and applying a simple, repeatable validation framework, the study aims to support labs in implementing standardized testing methodologies aligned with National Institute of Standards and Technology (NIST) and Scientific Working Group on Digital Evidence (SWGDE) recommendations [1], [2].

II. RELATED WORK AND BACKGROUND

Prior research and standards such as the NIST Computer Forensic Tool Testing (CFTT) program and SWGDE minimum requirements for testing tools put a heavy emphasis on accuracy, repeatability, and documentation of known limitations [1], [2]. Previous comparative studies have shown differences in how tools interpret timestamps, handle deleted file recovery, and parse file system structures [3], [4].

Autopsy is widely adopted due to its open-source nature, modular architecture, and support for multiple file systems (e.g., NTFS, FAT, EXT). However, discrepancies in file recovery and metadata interpretation have been reported in literature [4]. This study extends previous work by testing Autopsy with a controlled, reproducible dataset to measure both accuracy and repeatability.

III.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
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Number equations consecutively. Equation numbers, within parentheses, are to position flush right, as in (1), using a right tab stop. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate

exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Note that the equation is centered using a center tab stop. Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

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- The word “data” is plural, not singular.
- The subscript for the permeability of vacuum μ_0 , and other common scientific constants, is zero with subscript formatting, not a lowercase letter “o”.
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an “inset”, not an “insert”. The word alternatively is preferred to the word “alternately” (unless you really mean something that alternates).
- Do not use the word “essentially” to mean “approximately” or “effectively”.
- In your paper title, if the words “that uses” can accurately replace the word “using”, capitalize the “u”; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones “affect” and “effect”, “complement” and “compliment”, “discreet” and “discrete”, “principal” and “principle”.
- Do not confuse “imply” and “infer”.
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- There is no period after the “et” in the Latin abbreviation “et al.”.
- The abbreviation “i.e.” means “that is”, and the abbreviation “e.g.” means “for example”.

An excellent style manual for science writers is [7].

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A. Authors and Affiliations

The template is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

1) *For papers with more than six authors:* Add author names horizontally, moving to a third row if needed for more than 8 authors.

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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced. Styles named “Heading 1”, “Heading 2”, “Heading 3”, and “Heading 4” are prescribed.

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a) *Positioning Figures and Tables:* Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation “Fig. 1”, even at the beginning of a sentence.

TABLE I. TABLE TYPE STYLES

Table Head	Table Column Head		
	Table column subhead	Subhead	Subhead
copy	More table copy ^a		

Fig. 1. Example of a figure caption. (figure caption)

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

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REFERENCES

- [1] “Computer Forensics Tool Testing Program (CFTT),” NIST, <https://www.nist.gov/itl/ssd/software-quality-group/computer-forensics-tool-testing-program-cftt> (accessed Oct. 22, 2025).
- [2] “Publication documents,” SWGDE, <https://www.swgde.org/documents/published-complete-listing/> (accessed Oct. 22, 2025).
- [3]
- [4]
- [5]
- [6]

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